

To: PhD Admissions Committee — Computer Science and Engineering (DISI) **Institution:** University of Bologna (Alma Mater Studiorum) **Subject:** Motivation — PhD in Computer Science and Engineering (XLII / AY 2026-27) — Trustworthy / Responsible AI — Nauval Zulfikar **Ref:** [TODO: verify with user — round-2 call reference]

Dear Members of the Admissions Committee,

I am applying to the PhD in Computer Science and Engineering at DISI to pursue research on the **reliability and robustness of deployed machine-learning and LLM systems** — the technical core of trustworthy AI. I hold an MSc in Business Analytics from Aston University, awarded with First Class distinction, and my interest in this field is not abstract: it grew directly out of building and maintaining models that real institutions depended on.

At Bank Muamalat, a licensed bank, I developed NLP models for credit and risk scoring and was responsible for keeping them trustworthy after deployment — running regular validation cycles, monitoring model staleness, and producing audit-grade documentation for regulatory compliance. That experience taught me that the decisive question is rarely a model's accuracy at training time, but whether it stays reliable, robust, and accountable once it is live. This is the question I want to spend a PhD answering rigorously.

I bring methodological habits well suited to that aim. My Crowd-Shipping agent-based model was a 450-run controlled parameter sweep with literature-anchored, reproducible results — my closest experience to a controlled scientific experiment, and the discipline I would carry into trustworthy-AI evaluation. My LLM-Shipper-Profiles project, a retrieval-augmented system with a fully traceable rule-based fallback, was built deliberately to be reproducible and ablatable. My MSc dissertation fine-tuned DeBERTa-v3 transformers and evaluated emergent biases against literature baselines, and my toolkit includes causal inference and Bayesian methods for separating genuine drift from noise. Together these let me treat robustness and reliability as hypotheses to be tested under controlled, ablatable conditions rather than asserted.

I am drawn to Bologna specifically because DISI's strength in trustworthy, responsible, and distributed AI is the right setting for this work, where reliability and governance questions become sharper under distributed deployment. *[I would tailor my research project to a specific DISI group and supervisor — to be confirmed against the call.]* I would welcome the chance to contribute rigorous, reproducible evaluation methods to that effort.

I am committed to bridging from applied analytics toward the formal and systems depth of a CS&E PhD, and I am prepared to relocate and begin as scheduled.

Thank you for your consideration.

Sincerely, Nauval Zulfikar zulfikar.nauval1998@gmail.com · +44 7300 469048 · nauvalzulfikar.vercel.app

Grounded entirely in the applicant's verified profile. Round-2 reference, target group, and supervisor are marked for verification on the official call PDF.